

Assignment :1

Session: 1

❖ Name 5 IT job roles and write one line about what each role does.

1. Software Developer – Designs, writes, tests, and maintains computer programs and applications.
2. Network Administrator – Manages and maintains an organization's computer networks and internet connections.
3. Cybersecurity Analyst – Protects systems and data from cyber threats, attacks, and unauthorized access.
4. Database Administrator (DBA) – Organizes, manages, and secures databases to ensure data is available and accurate.
5. IT Support Technician – Helps users solve hardware, software, and technical problems.

❖ List 3 product companies and 3 service companies with one example of work each does.

Product Companies:

1. **Microsoft** – Develops software products like Windows operating system and Microsoft Office.
2. **Apple** – Designs and manufactures products such as the iPhone, iPad, and Mac computers.
3. **Adobe** – Creates software products like Photoshop, Illustrator, and Acrobat.

Service Companies:

- 1. Tata Consultancy Services (TCS)** – Provides IT consulting, software development, and business solutions for clients.
- 2. Infosys** – Offers software development, cloud services, and digital transformation solutions.
- 3. Wipro** – Delivers IT consulting, application development, and technology support services.

❖Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

GitHub: Git Hosting Platform; not an acronym.

- Used to store, manage, and collaborate on software code using Git version control.

Slack: Searchable Log of All Communication and Knowledge.

- Used for team communication, messaging, and collaboration.

Jira: Used to track projects, tasks, bugs, and software development work.

IDE: Integrated Development Environment.

- Used to write, edit, debug, and test computer programs.

API: application programming interface.

- Allows different software applications to communicate and share data with each other.

❖ **Name any 5 programming languages and mention one popular use case for each.**

Python: Data science, artificial intelligence, and web development.

Java: Building Android applications and enterprise software.

C++: developing games, operating system, and high-performance applications.

C#: developing window applications and game using unity.

JavaScript: creating interactive websites and web applications.

❖ **What is the difference between a developer and a tester? Answer in 2 lines.**

Developer: Writes, builds, and maintains software applications and features. They translate user needs and business requirements into functional code using programming languages like python, java, or JavaScript.

Tester: Checks the software for bugs, verifies it works correctly, and reports issues before release. They act as a safeguard by identifying bugs, preventing critical failures, and enhancing overall user satisfaction.

Session: 2

❖ **What are the three main layers of software architecture? Write one line about each.**

1. Presentation Layer (UI): Handles user interaction and displays information to the user.

2. Business Logic Layer (Application Layer) – Processes data and implements the application's rules and functionality.

3. Data Layer (Database Layer) – Stores, retrieves, and manages the application's data.

❖ **Define client and server in your own words with one real-life example.**

Client: A client is a device or application that requests services or data from another computer.

Server: A server is a computer or system that provides services or data to clients.

Real-life example: When you open YouTube on your phone, your phone is the client, and YouTube's computers that send the videos are the servers.

❖ **Write 2 differences each between desktop apps, web apps, and mobile apps.**

Desktop Apps:

1. Installed on a computer.
2. Can work without an internet connection.

Web Apps:

1. Run in a web browser.
2. Usually require an internet connection to use.

Mobile Apps:

1. Installed on smartphones or tablets.
2. Can use device features like the camera, GPS, and notifications.

❖ **Name any 3 databases and mention whether each is SQL or NoSQL.**

Database	type
MySQL	SQL
PostgreSQL	SQL
MongoDB	NoSQL

Session: 3

- ❖ Write the full form and one-line use of:
HTTP, HTTPS, FTP, SMTP.

HTTP: Hypertext Transfer Protocol.

- Used to transfer web pages and data between a web browser and a web server.

HTTPS: Hypertext Transfer Protocol Secure.

- Used to securely transfer data between a web browser and a web server using encryption.

FTP: File Transfer Protocol.

- Used to upload and download files between computers over a network.

SMTP: Simple Mail Transfer Protocol.

- Used to send emails from an email client to a mail sever or between mail server.

- ❖ What is the difference between IPv4 and IPv6? Answer in 2 lines.

IPv4: uses 32-bit addresses.

Ex: (192.168.1.1)

IPv6: uses 128-bit addresses.

Ex: (2001:db8::1)

IPv6 also offers improved efficiency, security features, and better support for modern internet devices compared to IPv4.

❖ **Run nslookup google.com on your system and write down the IP address you get.**

Server: 8.8.8.8

Address: 8.8.8.8#53

❖ **What do these HTTP status codes mean: 200, 301, 404, 500, 403?**

- **200 OK** – The request was successful, and the server returned the requested resource.
- **301 Moved Permanently** – The requested page has been permanently moved to a new URL.
- **404 Not Found** – The requested page or resource could not be found on the server.

- **500 Internal Server Error** – The server encountered an unexpected error while processing the request.
- **403 Forbidden** – The server understood the request but refuses to authorize access to the resource.

❖ **Explain DNS hierarchy (root, TLD, domain, subdomain) with `www.google.com` as an example.**

The **DNS (Domain Name System) hierarchy** is organized from the most general level to the most specific level.

Using **`www.google.com`** as an example:

- **Root (.)** – The top level of the DNS hierarchy. It is represented by a dot (.) at the end of a domain name (usually hidden), e.g., `www.google.com`.
- **TLD (Top-Level Domain)** – **`.com`** is the top-level domain.
- **Domain (Second-Level Domain)** – **`google`** is the registered domain name under `.com`.

- **Subdomain** – **www** is a subdomain of google.com, commonly used for the website.

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 - **Subdomain** – **www** is a subdomain of google.com, commonly used for the website.
-
- Root = .
 - TLD = com
 - Domain = google
 - Subdomain = www

Session: 4

❖ List all 6 phases of SDLC in order with one line about each.

1. **Planning** – Define the project goals, scope, resources, and timeline.

2. **Requirement Analysis** – Gather and analyse user needs and system requirements.

3. **Design** – Create the system architecture, database design, and user interface plans.

4. **Development (Coding)** – Programmers write and build the software according to the design.

5. **Testing** – Check the software for errors, bugs, and quality issues before release.

6. **Maintenance** – Update, fix, and improve the software after deployment.

❖ **Write 3 differences between Agile and Waterfall models.**

Agile Model:

- Development is iterative with frequent changes and releases.
- Customer feedback is collected continuously throughout the project.
- Testing is performed throughout development cycles.

Waterfall Model:

- Development follows a fixed sequence of phases with fewer changes.
- Customer feedback is mainly collected at the beginning and end.
- Testing is usually done after the development phase is completed.

❖ **Name the 3 main Scrum roles and write one line about the responsibility of each.**

1. **Product Owner** – Manages the product backlog and ensures the team delivers value according to customer needs.

2. **Scrum Master** – Facilitates Scrum processes, removes obstacles, and helps the team work effectively.

3. **Development Team** – Designs, builds, tests, and delivers the product increment during each sprint.

❖ **What is a Sprint in Scrum? Mention its typical duration.**

A Sprint in Scrum is a short, fixed period in which a Scrum team works to complete selected tasks and deliver a usable product increment. It helps teams organize work into smaller, manageable goals. During a Sprint, the team performs planning, development, testing, and review activities. The Sprint goal defines what the team aims to achieve within that period. A typical Sprint duration is 2 to 4 weeks, with 2 weeks being the most common duration.

❖ **What do these Jira board columns mean: To Do, In Progress, Done, Backlog?**

The meanings of common Jira board columns are:

1. Backlog – Contains all planned tasks, user stories, and issues that are not yet ready to be worked on.
2. To Do – Contains tasks that are selected for the current work cycle but have not started yet.
3. In Progress – Shows tasks that the team is currently working on.
4. Done – Contains completed tasks that have been finished, reviewed, and accepted.

Session: 5

❖ Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

- **</> bash**

git init

- **</> bash**

git add.

- **</> bash**

git commit -m "Your commit message"

- **</> bash**

git push origin main

- **</> bash**

git pull origin main

- ❖ **What is the difference between git fetch and git pull? Answer in 2 lines.**

Git fetch:

downloads the latest changes from a remote repository but does not merge them into your current branch.

Git pull:

downloads the changes and automatically merges them into your current branch.

- ❖ **Create a new GitHub repository, push any small text file from your local system, and share the repo link.**

git init

git add sample.txt

git commit -m "Add sample text file"

git branch -M main git remote add origin https://github.com/aryapatel2610/python_backend


```
git push -u origin main
```

Your repository link will be:

https://github.com/aryapatel2610/python_backend

❖ **What is the difference between a Fork and a Pull Request? Answer in 2 lines.**

Fork:

creates a personal copy of someone else's repository so you can make changes independently.

Pull:

Request is a request to merge your changes from your fork/branch into the original repository.

❖ **Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.**

- **Create and switch to a new branch**

git checkout -b feature-test

- **Create a small file:**

echo <"Feature test" > feature.txt

- **Stage and commit changes**

git add feature.txt

git commit -m "Add feature test file"

- **Switch back to main branch**

git checkout main

- **Merge feature branch into main**

git merge feature-test





